

1 Final-Test Attack

Lecturer priorities: worksheets IP1–IP5; z-transform lookup/inversion; review problems: shifted convolution, delayed-sum magnitude/phase, separable-kernel proof/count. Show formula → substitution → units/ROC/padding → interpretation. Use atan2; convolution flips, correlation does not.

2 DSP I: Signals and LTI Systems

2.1 Signal Essentials

$x[n] = \sum_k x[k]\delta[n - k]$; $u[n] = 1$ for $n \geq 0$. Shift $x[n - n_0]$ right; fold $x[-n]$. Periodic iff $x[n + N] = x[n]$; $e^{j\omega_0 n}$ iff $\omega_0/(2\pi)$ rational.

$$E = \sum_n |x[n]|^2, \quad P = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{-N}^N |x[n]|^2, \quad x_{e/o}[n] = \tfrac{1}{2}(x[n] \pm x[-n])$$

2.2 System Tests and Convolution

Linearity: $T\{ax_1 + bx_2\} = aT\{x_1\} + bT\{x_2\}$. TI: shifting input by n_0 shifts output by n_0 . Causal: output at n uses no $x[k]$, $k > n$. BIBO: bounded input $\Rightarrow$ bounded output.

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

For LTI: causal iff $h[n] = 0$ for $n < 0$; stable iff $\sum_n |h[n]| < \infty$. Convolution is commutative, associative, distributive; $x * \delta = x$; $x * \delta[n - n_0] = x[n - n_0]$.

Finite supports: if x starts/ends at n_x^-, n_x^+ and h at n_h^-, n_h^+ , then y spans n_x^-, n_h^+ through $n_x^+ + n_h^+$ and $L_y = L_x + L_h - 1$.

2.3 [REVIEW] Shifted-Step Convolution

Given $h[n] = a^n u[n]$, $|a| < 1$:

$$s[n] = (u * h)[n] = \sum_{r=0}^n a^r = \frac{1 - a^{n+1}}{1 - a} u[n]$$

For $x[n] = u[n + 5] - u[n - 10]$, define $t[n] = (x * h)[n]$:

$$\begin{aligned} t[n] &= s[n + 5] - s[n - 10] \\ t[n] &= \frac{1 - a^{n+6}}{1 - a} u[n + 5] - \frac{1 - a^{n-9}}{1 - a} u[n - 10] \end{aligned}$$

Two branches, upper added and delayed-by-2 lower subtracted:

$$\begin{aligned} y[n] &= x[n] * h[n] - x[n - 2] * h[n] = t[n] - t[n - 2] \\ t[n - 2] &= \frac{1 - a^{n+4}}{1 - a} u[n + 3] - \frac{1 - a^{n-11}}{1 - a} u[n - 12] \end{aligned}$$

Do not confuse shifting one convolution input with replacing only its variable: $(x[n - n_0] * h[n]) = (x * h)[n - n_0]$.

2.4 Difference Equations

For zero initial conditions,

$$y[n] + \sum_{k=1}^N a_k y[n - k] = \sum_{k=0}^M b_k x[n - k]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

FIR: finite $h[n]$, usually nonrecursive. IIR: infinite $h[n]$, usually recursive.

3 DSP II: z-Transform and DTFT

3.1 z-Transform Quick-Lookup Table

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}, \quad q = z^{-1}, \quad D_a = 1 - aq$$

Sequence $x[n]$	Transform $X(z)$	ROC
$\delta[n]$	1	all z
$\delta[n - k]$	z^{-k}	$z \neq 0$ if $k > 0$
$u[n]$	$1/(1 - q)$	$ z > 1$
$u[n - k]$	$z^{-k}/(1 - q)$	$ z > 1$
$a^n u[n]$	$1/D_a$	$ z > a $
$-a^n u[-n - 1]$	$1/D_a$	$ z < a $
$a^n u[-n - 1]$	$-1/D_a$	$ z < a $
$na^n u[n]$	aq/D_a^2	$ z > a $
$-na^n u[-n - 1]$	aq/D_a^2	$ z < a $
$(n + 1)a^n u[n]$	$1/D_a^2$	$ z > a $
$a^n[u[n] - u[n - N]]$	$\frac{1 - (aq)^N}{1 - aq}$	$N > 1: z \neq 0$

Let $D_{a,\omega} = 1 - 2a \cos \omega q + a^2 q^2$:

$$\begin{aligned} a^n \cos(\omega_0 n) u[n] &\leftrightarrow \frac{1 - a \cos \omega_0 q}{D_{a,\omega_0}}, & |z| > |a|, \\ a^n \sin(\omega_0 n) u[n] &\leftrightarrow \frac{a \sin \omega_0 q}{D_{a,\omega_0}}, & |z| > |a|. \end{aligned}$$

Set $a = 1$ for undamped sine/cosine. Repeated pole $(1 - pq)^{-m}$ produces a degree- $(m - 1)$ polynomial in n times p^n .

3.2 z-Transform Properties and ROC

$$\begin{aligned} x[n - k] &\leftrightarrow z^{-k} X(z) & c^n x[n] &\leftrightarrow X(c^{-1} z) \\ x[-n] &\leftrightarrow X(z^{-1}) & nx[n] &\leftrightarrow -z dX/dz \\ x_1 * x_2 &\leftrightarrow X_1 X_2 & x^*[n] &\leftrightarrow X^*(z^*) \end{aligned}$$

Initial value for causal x : $x[0] = \lim_{z \rightarrow \infty} X(z)$. DTFT is $X(e^{j\omega})$ only when the unit circle lies in the ROC.

ROC excludes poles. Right-sided: outside outermost pole; left-sided: inside innermost pole; two-sided: annulus. Causal rational: ROC includes infinity. Stable: ROC includes $|z| = 1$. Causal rational stable: every uncanceled pole is strictly inside the unit circle. Same algebraic $X(z)$ can represent different sequences; **ROC decides sidedness**.

3.3 Inverse z-Transform Attack

Workflow: factor/cancel removable factors $\rightarrow$ make proper $\rightarrow$ partial fractions or power series $\rightarrow$ use ROC to assign sidedness. Polynomial terms $c_k z^{-k}$ become $c_k \delta[n - k]$.

$$X(z) = P(z^{-1}) + \sum_k \frac{A_k}{1 - p_k z^{-1}}, \quad A_k = \left. \frac{(z - p_k)X(z)}{z} \right|_{z=p_k}$$

Residue shortcut above: distinct simple poles; repeated poles need repeated-pole partial fractions/derivatives.

$$\frac{A}{1 - pz^{-1}} \leftrightarrow \begin{cases} A p^n u[n], & |z| > |p| \\ -A p^n u[-n - 1], & |z| < |p| \end{cases}$$

Exterior ROC: all pole terms right-sided. Interior ROC: all left-sided. Annular ROC: poles inside the annulus's inner boundary give right-sided terms; poles outside its outer boundary give left-sided terms. For a *right-sided* conjugate pair $p = re^{j\omega_0}$, combine residues into $2|A|r^n \cos(\omega_0 n + \angle A) u[n]$.

3.4 DTFT and Properties

$$X(\omega) = \sum_n x[n]e^{-j\omega n}, \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega)e^{j\omega n} d\omega$$

$X(\omega)$ is 2π -periodic; it is continuous when $\sum_n |x[n]| < \infty$. Useful pairs/properties:

$$a^n u[n] \leftrightarrow \frac{1}{1 - ae^{-j\omega}}, \quad |a| < 1; \quad x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(\omega)$$

$$e^{j\omega_0 n} x[n] \leftrightarrow X(\omega - \omega_0), \quad nx[n] \leftrightarrow j \frac{dX}{d\omega}, \quad x * h \leftrightarrow XH$$

Real $x[n]$: $X(\omega) = X^*(-\omega)$; $|X|$, $\Re X$ even; $\Im X$ and phase odd (phase modulo 2π , away from zeros).

$$\sum_n |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$$

Length- N rectangle, $0 \leq n \leq N - 1$:

$$X(\omega) = e^{-j\omega(N-1)/2} \frac{\sin(N\omega/2)}{\sin(\omega/2)}$$

At $\omega = 2\pi k$, use the removable-limit value $X = N$.

3.5 [REVIEW] Magnitude and Phase

For $y[n] = x[n] + x[n - L]$:

$$H(\omega) = 1 + e^{-jL\omega} = 2 \cos(L\omega/2) e^{-jL\omega/2}$$

$$|H| = 2|\cos(L\omega/2)|, \quad \angle H = \begin{cases} -L\omega/2, & \cos(L\omega/2) > 0 \\ -L\omega/2 + \pi, & \cos(L\omega/2) < 0 \end{cases}$$

(phase modulo 2π ; undefined at zeros). For the review $L = 4$:

$$H = 2 \cos(2\omega) e^{-j2\omega}, \quad |H| = 2|\cos 2\omega|, \quad \angle H = -2\omega \text{ (+}\pi \text{ if } \cos 2\omega < 0)$$

General: $|A/B| = |A|/|B|$, $\angle(A/B) = \angle A - \angle B$; $|e^{-j\omega n_0}| = 1$, phase = $-\omega n_0$.

First-order smoother template:

$$y[n] = ay[n - 1] + bx[n] \Rightarrow H(\omega) = \frac{b}{1 - ae^{-j\omega}}$$

$$|H| = \frac{|b|}{\sqrt{1 + a^2 - 2a \cos \omega}}, \quad \angle H = \angle b - \text{atan2}(a \sin \omega, 1 - a \cos \omega)$$

For unity DC gain choose $b = 1 - a$; stability requires $|a| < 1$.

3.6 Sinusoidal Eigenfunction and Sampling

$$x[n] = A \cos(\omega_0 n + \phi) \Rightarrow y[n] = A|H(\omega_0)| \cos(\omega_0 n + \phi + \angle H(\omega_0))$$

$$F_s = 1/T, \quad \omega = 2\pi F/F_s, \quad F = \omega F_s/(2\pi)$$

Nyquist: $F_s > 2B$ (endpoint equality requires special assumptions). Fold aliases into $[-F_s/2, F_s/2]$ using $F_a = F - kF_s$.

4 DSP III: Filters, DFT, DCT, FFT

4.1 Pole-Zero Filter Reading

$H(\omega) = H(z)|_{z=e^{j\omega}}$, $|H| = |G| \prod |e^{j\omega} - z_k| / \prod |e^{j\omega} - p_k|$. Unit-circle zero: null; nearby pole: resonance; radius near 1: narrower/longer decay. Real coefficients need conjugate pairs; $\tau_g = -d\angle H/d\omega$. Comb $1 + gz^{-D}$ gives periodic minima (exact nulls if $|g| = 1$); all-pass changes phase only.

4.2 DFT, Circular Convolution, FFT

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}, \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{j2\pi kn/N}$$

$W_N = e^{-j2\pi/N}$; $X[k] = \sum_n x[n]W_N^{kn}$. DFT multiplication $\leftrightarrow$ N -point circular convolution. For linear convolution zero-pad to $N \geq L_x + L_h - 1$.

$$X[k] = E[k] + W_N^k O[k], \quad X[k + N/2] = E[k] - W_N^k O[k]$$

Radix-2 FFT: $N = 2^m$; direct DFT N^2 multiplies vs FFT $(N/2) \log_2 N$. Leakage: rectangular window has narrow main lobe/high sidelobes; Hann reverses that tradeoff. STFT gives local spectra; DCT is real and energy-compacting for compression.

5 DIP I: Fundamentals and Transforms

5.1 Sampling, Quantization, Resolution

$M \times N$, k bpp: storage MNk bits and $L = 2^k$ levels. Sampling discretizes space; quantization discretizes amplitude. Cones: bright/color; rods: dim. Enhancement subjective; restoration model-based. Affine preserves parallelism; projective preserves lines. Weber $\Delta I_c/I$: smaller is better.

5.2 Optics and Pixel Geometry

Thin lens and magnification:

1/f = 1/d_o + 1/d_i, m = h_i/h_o = -d_i/d_o

d_o = f d_i / (d_i - f). IP1: h_o = 2 mm, d_o = 5 mm, d_i = 17 mm => |h_i| = 6.8 mm, f = 16.94 mm; not within 1.5-mm fovea. Match units; m < 0 means inversion. For pixels p = (x, y), q = (s, t):

D_E = sqrt((x-s)^2 + (y-t)^2), D_4 = |x-s| + |y-t|, D_8 = max(|x-s|, |y-t|), D_p = (|x-s|^p + |y-t|^p)^(1/p)

IP1: (3, 4) -> (7, 1): D_E = 5, D_4 = 7, D_8 = 4, D_3 = 3.91 = 4.498. N_4: horizontal/vertical; N_D: diagonals; N_8 = N_4 union N_D. Adjacency: both values in V -> check geometry; a diagonal is m-adjacent only if common N_4 contains no pixel in V. 5.3 Point and Geometric Transforms Negative: s = L - 1 - r. Log: s = c ln(1 + r) expands dark levels (IP2 uses c = 45). Gamma: s = cr^gamma; gamma > 1 brightens, gamma < 1 darkens. Threshold: s = 0 if r < T, else L - 1. Evaluate -> round as stated -> clip to [0, L - 1]. Homogeneous coordinates (column-vector convention):

[x'; y'; 1] = [1 0 tx; 0 1 ty; 0 0 1] [cos theta sin theta 0; sin theta cos theta 0; 0 0 1] [sx 0 0; 0 sy 0; 0 0 1] [x; y; 1]

Rightmost acts first; inverse mapping avoids holes. Nearest: fast/blocky; bilinear: smooth; bicubic: sharper. IP2: 45 deg CCW, scale 2, translate (3, 4): H = TSR maps (2, 3) -> (1.586, 11.071). Declare (x, y) vs row/column; geometry moves locations, not ideal intensity. 5.4 Histograms h(r_k) = n_k, normalized p(r_k) = n_k / (MN).

s_k = Q((L - 1) * sum_{j=0}^k p(r_j)), Q = round or floor as specified

HE spreads intensities but may amplify noise. IP2: 2-bit counts (80, 800, 0, 20) give CDF (.0889, .9778, .9778, 1); rounding maps (0, 1, 2, 3) -> (0, 3, 3, 3).

6 DIP II: Spatial Filtering

6.1 Correlation, Convolution, Noise

g(x, y) = sum_{s=-a}^a sum_{t=-b}^b w(s, t) f(x + s, y + t) (correlation)

Convolution uses f(x - s, y - t), equivalently rotate w by 180 deg. Symmetric kernel: same result. Smoothing kernels usually sum to 1; derivative kernels sum to 0. Same-size odd kernel: pad by half-width/height.

Hand workflow: decide full/same/valid output and padding -> flip kernel only for convolution -> center at target -> elementwise multiply and sum. The IP3 symmetric kernel [1 2 1; 2 4 2; 1 2 1] gives identical correlation and convolution; a nonsymmetric kernel does not.

w = [1 2 1; 0 4 1; 0 0 2], w_flip = [2 0 0; 1 4 0; 1 2 1]

Box mean reduces random noise but blurs edges. Gaussian:

G(s, t) = K exp[-(s^2 + t^2) / (2 sigma^2)], h = G / sum G

Larger sigma / kernel => stronger smoothing. Median: sort neighborhood and select middle; best for salt-and-pepper and preserves edges better. IP3 Gaussian example, sigma = 2:

G = [.7788 .8825 .7788; .8825 1 .8825; .7788 .8825 .7788], sum G = 7.6452, h = G / 7.6452

Thus corner/edge/center normalized weights are .1019/.1154/.1308. Dotting h with neighborhood [100 102 98; 101 99 103; 97 100 102] gives 100.26 approx 100. Compute unique values by symmetry -> normalize -> weighted sum.

6.2 [REVIEW] Separable Kernels

A nonzero w is separable iff w = cr (outer product), equivalently rank 1: every row/column is a scalar multiple (all 2 x 2 minors zero). The zero kernel has rank 0 and is also expressible as an outer product. Then

w * f = c * (r * f)

For image M x N, kernel m x n:

C_2D = MNmn, C_sep = MN(m + n), S = mn / (m + n)

Square m x m: S = m/2, reducing O(MNmn^2) to O(MNm). For desired speed-up S_0:

m/2 >= S_0 => m >= 2S_0; thus 10x => m >= 20

Proof by sums: if w(s, t) = c(s)r(t),

g = sum_s c(s) [sum_t r(t) f(x - s, y - t)]

which is a 1-D row pass followed by a 1-D column pass. Ignore border overhead only when using the theoretical count.

Extract factors (IP3): choose nonzero pivot E = w_ij; let c be pivot column and r be pivot row; then test w = c(r/E).

[1 3 1; 2 6 2] = [1 1; 2 2] [1 3 1]

For 512 x 512 and separable 5 x 5: direct = 512^2(25) = 6,553,600; separable = 512^2(10) = 2,621,440; speed-up = 2.5x.

6.3 Sharpening and Edges

Four-neighbor Laplacian:

nabla^2 f = f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1) - 4f(x, y)

Use g = f + c nabla^2 f with sign matching the chosen Laplacian kernel. Unsharp/high-boost:

m = f - f_bar, g = f + km = (1 + k)f - k f_bar

k = 1: unsharp; k > 1: high-boost. Sobel:

G_x = [-1 0 1; -2 0 2; -1 0 1], G_y = G_x^T, |nabla f| approx |g_x| + |g_y|

IP3 checks: median neighborhood [50, 50, 50; 50, 200, 50; 60, 80, 50] has median 50. Rounded worksheet Gaussian gives f_bar = 89.982, k = 1.5, g = 90.027 approx 90; exact normalized symmetry gives 90.

7 DIP III: Frequency-Domain Filtering

7.1 2-D Sampling and DFT

Nyquist independently in each direction:

1/Delta x > 2 nu_max, 1/Delta y > 2 nu_max

IP4 example: maxima (60, 45) Hz require (f_sx, f_sy) > (120, 90) Hz; sampling at (100, 80) aliases in both directions. Frequency-bin spacings are Delta u = 1/(M Delta x), Delta v = 1/(N Delta y).

F(u, v) = sum_{x=0}^{M-1} sum_{y=0}^{N-1} f(x, y) e^{-j 2 pi (ux/M + vy/N)}

f(x, y) = 1/(MN) sum_{u=0}^{M-1} sum_{v=0}^{N-1} F(u, v) e^{j 2 pi (ux/M + vy/N)}

|F| = sqrt((Re F)^2 + (Im F)^2), phi = atan2(Im F, Re F), F(0, 0) = MN f_bar

Magnitude gives strength; phase carries structure. Translation changes phase only; (-1)^{x+y} centers DC. Real image: F(M - u, N - v) = F*(u, v).

7.2 Convolution, Padding, Workflow

DFT multiplication gives circular convolution. For A x B image and C x D kernel, linear convolution requires

P >= A + C - 1, Q >= B + D - 1

Equal-size shortcut: P = 2M, Q = 2N. IP4: 128^2 * 64^2 needs minimum 191^2.

Workflow: pad -> multiply (-1)^{x+y} -> DFT F -> construct centered H -> G = HF -> IDFT -> multiply (-1)^{x+y} -> real part -> crop. Padding prevents wraparound; centering puts radial-filter origin at the middle.

7.3 Radial Filters

On padded P x Q grid:

D(u, v) = sqrt((u - P/2)^2 + (v - Q/2)^2), H_ILP = {1 D <= D_0; 0 D > D_0}, H_GLP = e^{-D^2 / (2 D_0^2)}, H_BLP = 1 / (1 + (D/D_0)^{2n})

High-pass H_HP = 1 - H_LP; set center to 0. Ideal: most ringing; Gaussian: least; higher Butterworth n: sharper/more ringing. Smaller LPF D_0: more smoothing.

Butterworth (course padding convention): pad 256^2 -> 512^2, center (256, 256); at (60, 60), D = 277.19 and D_0 = 50, n = 2 => H approx 0.0011.

Costs: spatial MNmn; padded FFT approx 2PQ log2(PQ) + PQ. IP4 rough 2048^2, 201^2: 918x FFT speed-up (ignores padding/constants). Prefer FFT for large nonseparable kernels.

8 DIP IV: Image Compression (Worksheet IP5)

8.1 Compression Ratio and Redundancy

For original size b_raw and compressed size b_c:

C_R = b_raw / b_c, R_D = 1 - 1/C_R = 1 - b_c / b_raw

Compression removes coding, interpixel/spatial, and psychovisual redundancy. Lossless methods recover exact data; lossy methods discard information for higher compression.

IP5 Q1: 4656 x 3492 24-bit RGB has

b_raw = 4656(3492)(24) = 390,210,048 bits = 48.776256 MB

For a 5.5323 MB JPEG, C_R = 8.8166 : 1 and R_D = 88.66% using decimal MB. State whether MB or MiB is used.

8.2 Code Length and Entropy

For symbols with probabilities p_i and code lengths l_i:

L_avg = sum_i p_i l_i, b_coded = N L_avg, H = - sum_i p_i log2 p_i

H <= L_avg < H + 1 for an optimal binary prefix code. For raw b bits/pixel, the zero-memory upper compression limit is C_R,max = b/H. A fixed code for q symbols needs floor(log2 q) bits/symbol.

IP5 Q2/Q4: for p = (.25, .47, .25, .03), four symbols need 2 fixed bits, so 8-bit raw gives 4 : 1. With code lengths (2, 1, 3, 3):

L_avg = .25(2) + .47(1) + .25(3) + .03(3) = 1.81, C_R = 8/1.81 = 4.42 : 1

H approx 1.664 bpp, C_R,max = 8/H approx 4.81 : 1

8.3 Run-Length Coding (RLE)

Encode each run as (value, length). If value uses b_v bits and length l_l bits:

b_RLE = (# runs)(b_v + l_l)

Best for long constant runs; poor for noisy/rapidly varying data. IP5 Q3: a 256 x 255 8-bit image with each row constant has raw 256(255)(8) = 522,240 bits. One 8-bit value + 8-bit run length per row uses 256(16) = 4096 bits, so C_R = 127.5 : 1, R_D = 99.216%.

8.4 Huffman Coding

Build: repeatedly merge the two smallest probabilities -> label branches 0/1 -> read root-to-leaf codes. Codes are prefix-free; ties can produce different but equally valid codes. Decode one prefix at a time using the exact table built.

For p(a_1, ..., a_6) = (.10, .40, .06, .10, .04, .30), one valid table is

a_2 = 0, a_6 = 10, a_5 = 1100, a_3 = 1101, a_1 = 1110, a_4 = 1111

with L_avg = 2.2 bits/symbol. Then

010100111100 = 0|10|10|0|1111|0|0 -> a_2 a_6 a_6 a_2 a_4 a_2 a_2

8.5 Arithmetic Coding

Assign cumulative source intervals [C_low(a), C_high(a)). Starting from [L, U) with width W = U - L:

L' = L + W C_low(a), U' = L + W C_high(a)

Repeat per symbol; decode with known length/end symbol. For a_1 a_2 a_3 a_4:

Q5 six-symbol model: [.03128064, .03129504], tag .03129; textbook p = (.2, .2, .4, .2) : [.06752, .0688], tag .068.

9 DIP V: Color Essentials

RGB additive/display; CMY(K) subtractive/print; HSI separates intensity.

C = 1 - R, M = 1 - G, Y = 1 - B; K = min(C, M, Y)

If K not= 1: C' = (C - K)/(1 - K), similarly M', Y'; if K = 1, set C' = M' = Y' = 0. 24-bit RGB: 2^24 colors; R = G = B gray.

I = (R + G + B) / 3, S = 1 - 3 min(R, G, B) / (R + G + B)

theta = cos^-1 (1/2 [(R - G) + (R - B)] / sqrt((R - G)^2 + (R - B)(G - B))), H = { theta B <= G; 360 deg - theta B > G }

For grayscale S = 0, hue undefined. For brightness/HE/smoothing, process HSI I to reduce hue distortion; separate nonlinear RGB mappings can shift hue. Complement (1 - R, 1 - G, 1 - B); pseudocolor assigns colors to grayscale.